Combining Spectral-Domain OCT and Air-Puff Tonometry Analysis to Diagnose Keratoconus

Nan-Ji Lu, MSc, MD; Ahmed Elsheikh, PhD; Jos J. Rozema, MSc, PhD; Nikki Hafezi, MAS IP ETHZ; Ioannis M. Aslanides, MD, PhD, FRCOphth; Mark Hillen, PhD; Daniel Eckert, MSc; Christian Funck, MSc; Carina Koppen, MD, PhD; Le-Le Cui, MD, PhD; Farhad Hafezi, MD, PhD

ABSTRACT

PURPOSE: To investigate the diagnostic capacity of spectral-domain optical coherence tomography (SD-OCT) combined with air-puff tonometry using artificial intelligence (AI) in differentiating between normal and keratoconic eyes.

METHODS: Patients who had either undergone uneventful laser vision correction with at least 3 years of stable follow-up or those who had forme fruste keratoconus (FFKC), early keratoconus (EKC), or advanced keratoconus (AKC) were included. SD-OCT and biomechanical information from air-puff tonometry was divided into training and validation sets. AI models based on random forest or neural networks were trained to distinguish eyes with FFKC from normal eyes. Model accuracy was independently tested in eyes with FFKC and normal eyes. Receiver operating characteristic (ROC) curves were generated to determine area under the curve (AUC), sensitivity, and specificity values.

RESULTS: A total of 223 normal eyes from 223 patients, 69 FFKC eyes from 69 patients, 72 EKC eyes from 72 patients, and 258 AKC eyes from 258 patients were included. The top AUC ROC values (normal eyes compared with AKC and EKC) were Pentacam Random Forest Index (AUC = 0.985 and 0.958), Tomographic and Biomechanical Index (AUC = 0.983 and 0.925), and Belin-Ambrósio Enhanced Ectasia Total Deviation Index (AUC = 0.981 and 0.922). When SD-OCT and air-puff tonometry data were combined, the random forest AI model provided the highest accuracy with 99% AUC for FFKC (75% sensitivity; 94.74% specificity).

CONCLUSIONS: Currently, AI parameters accurately diagnose AKC and EKC, but have a limited ability to diagnose FFKC. AI-assisted diagnostic technology that uses both SD-OCT and air-puff tonometry may overcome this limitation, leading to improved treatment of patients with keratoconus.

[*J Refract Surg*. 2022;38(6):374-380.]

Keratoconus is a relatively common bilateral corneal ectasia disease,[1] characterized by local biomechanical weakness with possible asymmetric binocular involvement. As the disease develops, the weakening cornea becomes increasingly unable to resist the distension caused by intraocular pressure (IOP). This may lead to the development of a cone-shaped protrusion that causes increasing myopia and irregu-

From the Eye Hospital and School of Ophthalmology and Optometry, Wenzhou Medical University, National Clinical Research Center for Ocular Diseases, Wenzhou, Zhejiang, China (N-JL, IMA, L-LC, FH); Faculty of Medicine and Health Sciences, University of Antwerp, Wilrijk, Belgium (N-JL, JJR, NH, CK); the Department of Ophthalmology, Antwerp University Hospital, Edegem, Belgium (N-JL, JJR, CK); ELZA Institute, Dietikon, Switzerland (N-JL, NH, FH); Emmetropia Mediterranean Eye Institute, Heraklion, Greece (N-JL, IMA); School of Engineering, University of Liverpool, Liverpool, United Kingdom (AE); NIHR Biomedical Research Centre for Ophthalmology, Moorfields Eye Hospital NHS Foundation Trust and UCL Institute of Ophthalmology, London, United Kingdom (AE); Beijing Advanced Innovation Centre for Biomedical Engineering, Beihang University, Beijing, China (AE); Laboratory of Ocular Cell Biology, Center for Applied Biotechnology and Molecular Medicine, University of Zurich, Zurich, Switzerland (DE, CF, FH); Faculty of Medicine, University of Geneva, Geneva, Switzerland (FH); and USC Roski Eye Institute, University of Southern California, Los Angeles, California (FH).

Submitted: October 12, 2021; Accepted: April 13, 2022

Supported by Zhejiang Provincial Natural Science Foundation of China (Grant No. LY22H120007) and a Chinese Scholarship Council award (Grant No. 202008330323) (N-JL).

Disclosure: The authors have no financial or proprietary interest in the materials presented herein.

Correspondence: Le-Le Cui, MD, PhD, Eye Hospital and School of Ophthalmology and Optometry, Wenzhou Medical University, National Clinical Research Center for Ocular Diseases, 270 West Xueyuan Road, Wenzhou, Zhejiang, China. Email: cll@eye.ac.cn

doi:10.3928/1081597X-20220414-02

lar astigmatism. Early screening for keratoconus is important for laser vision correction to prevent triggering the keratoconus pathological process in susceptible patients. In addition, the timely diagnosis with close follow-up enables the early application of corneal cross-linking to halt disease progression and vision loss.

There are currently three types of non-contact clinical instruments used to diagnose keratoconus. The first type uses the Placido or Scheimpflug techniques to obtain shape parameters that include corneal curvature, thickness, and surface elevations. The second is anterior segment optical coherence tomography (AS-OCT), which provides measures of corneal epithelium thickness and corneal curvature and thickness.[2] The third is air-puff tonometry, which measures the pathological impact keratoconus has on corneal biomechanical integrity. To date, most researchers have applied these methods separately to diagnose early-stage keratoconus.

Although each instrument is continually optimized through software updates, the diagnostic efficiency of each individual instrument is limited by its technical specifications. Because these methods are complementary, it is important to consider a clinical approach that combines the contributions of multiple instrument types. To this end, Hwang et al[3] combined AS-OCT with Scheimpflug-based tomography to detect highly asymmetric keratoconus. However, this combination was restricted to structural diagnoses, which may not enable the full spectrum of patients with keratoconus to be diagnosed. Indeed, others have shown using air-puff technology that corneal biomechanical strength deteriorates in patients with keratoconus, before evidence of topographic anomalies can be observed.[4,5] In line with this finding, Ambrósio et al[6] combined Scheimpflug-based tomography with an air-puff device to detect subclinical ectasia, but the study did not record any information on the state of the epithelium. Finally, irregular epithelial profiles can influence tomographic measurements,[7,8] further highlighting the utility of using complementary techniques.

Inspired by these examples, the current study explored the diagnostic capability of combining AS-OCT with air-puff tonometry to differentiate between normal and keratoconic corneas. If successful, the combination may provide an improved approach to diagnose early-stage (EKC) and forme fruste (FFKC) keratoconus.

PATIENTS AND METHODS

This diagnostic study was conducted with the approval of the Ethics Committee of the Eye Hospital of Wenzhou Medical University. The study adhered to the tenets of the Declaration of Helsinki and its statement of ethical principles guiding the conduct of medical research involving human subjects. All participants signed informed consent.

Patient Groups

This study considered four groups of volunteers: a normal/control group, a group with FFKC, a group with EKC, and a group with advanced keratoconus (AKC), each defined according to the criteria described below. Given the large correlation between the fellow eyes of healthy participants, one randomly selected eye was included per person in the normal group. For the FFKC and EKC groups, only the eye that met the inclusion criteria was analyzed. The exception was in the AKC group where the contralateral eye from the patients with FFKC and EKC was included if they met the AKC group criteria.

The volunteers in the normal group were recruited from patients presenting for laser vision correction in the Eye Hospital of Wenzhou Medical University, using the inclusion criteria of no ocular or systemic abnormalities, no ocular surgery history, a stable corrected distance visual acuity (CDVA) of 20/20 or better for 2 years before surgery, and a 3-year follow-up after laser vision correction to exclude those with no clinical or tomographic signs of iatrogenic ectasia.[9]

The diagnosis of keratoconus required at least one slit-lamp finding (Fleischer ring, Vogt striae, or central thinning) and two signs of keratoconus on Scheimpflug topography (Pentacam HR; Oculus Optikgeräte GmbH), such as decreased thinnest pachymetry, skewed asymmetric bowtie/inferior steep or increased inferior steepness.

For the FFKC group, the inclusion criteria were: (1) the contralateral eye was diagnosed as having keratoconus according to the criteria above, (2) CDVA of 20/20 or better, (3) no keratoconus signs on slit-lamp examination, (4) maximum keratometry (Kmax) less than 47.40 diopters (D), (5) thinnest pachymetry of 480 µm or greater obtained by Pentacam HR, and (6) "normal" topography with the difference between the Kmax values in the inferior and superior areas at 3 mm (I-S value) less than 1.40 D, no skewed asymmetric bowtie/inferior steep, and keratoconus percentage index (KISA%) less than 60.

The inclusion criteria for the EKC group were based on severity of 1 in the Amsler-Krumeich classification of keratoconus: (1) Kmax less than 48.50 D and smallest thickness greater than 480 µm, (2) CDVA of 16/20 or better, and (3) no central scars and fewer than two slit-lamp findings.

Finally, the AKC group included all of those keratoconic eyes with parameters exceeding the criteria of the FFKC and EKC groups (ie, Kmax of 48.50 D or greater, smallest thickness less than 480 µm, and CDVA worse than 16/20, with at least one slit-lamp finding).

Data Acquisition and Evaluated Parameters

Participants were asked to discontinue wearing soft contact lenses for at least 2 weeks before the examination, or at least 4 weeks for rigid gas-permeable contact lenses.

Pentacam HR Scheimpflug Topography Data

To classify the patients, tomography measurements were obtained with the Pentacam HR (software version 1.21r59). Only eyes with a good quality score were considered, using the following parameters to assign the eyes to a severity group: steep keratometry, flat keratometry, Kmax, thinnest pachymetry, the difference between average inferior and superior corneal powers within 3 mm from the corneal center (IS-Value), and two artificial intelligence (AI) parameters: the Belin-Ambrósio Enhanced Ectasia Total Deviation Index (BAD-D) and the Pentacam Random Forest Index (PRFI). The Pentacam data were only used for the initial classification and were not included in the feature selection or AI training.

RTVue-XR Spectral-Domain OCT Data

First, a measurement was performed with the RTVue-XR Spectral-Domain OCT (Optovue, Inc), which is known to provide repeatable thickness maps of anomalous corneas.[10] This provided thickness maps for the whole cornea, the corneal epithelium, and stroma in the central (2-mm diameter), paracentral (2- to 5-mm), midperipheral (5- to 7-mm), and peripheral (7- to 9-mm) regions. In the latter three regions, the thickness was monitored in eight equally spaced points along the median circumference of the region, including the temporal, superior-temporal, superior, superior-nasal, nasal, inferior-nasal, inferior, and inferior-temporal positions.

Corvis ST Biomechanical Data

Finally, Corvis ST (software version 1.21r59; Oculus Optikgeräte GmbH) measurements were obtained, which recorded 41 parameters in two categories: independent parameters, such as IOP, biomechanically corrected IOP (bIOP), pachymetry, and 35 dynamic corneal response (DCR) parameters. The latter group included the ratio between the central deformation and the average of peripheral deformation at either 1 or 2 mm from center (DA ratio 1 mm/2 mm), peripheral corneal thickness increase (Pachy Slope), the Ambrósio's relational thickness to the horizontal profile (ARTh), the reciprocal of the radius during the concave state of the cornea (integrated radius), and the stiffness parameter at first applanation (SPA1).[11] Three other parameters were also recorded: the stress-strain index (SSI) and two AI parameters: the Corvis Biomechanical Index (CBI) developed from DCR parameters and the Tomographic and Biomechanical Index (TBI) developed from DCR and topography parameters. Only measurements with a good quality score were considered for analysis.

Artificial Intelligence

We used R software (version 4.0.4; R Foundation for Statistical Computing, Vienna, Austria) to develop two AI models based on random forest (RF) and neural networks (NN).

In the current study, before the AI models were trained based on SD-OCT and/or Corvis ST, feature selection was required among the parameters exported from SD-OCT and Corvis ST using the Boruta package (version 7.0.0)[12] for two reasons: (1) surgeons often prefer the use of minimal-optimal parameters for keratoconus diagnosis and (2) large features slow down AI models' algorithms, particularly in NN, and will simultaneously decrease the models' best possible performance.[13]

Models

Following feature selection, RF and NN models were developed based on the selected features from SD-OCT and/or Corvis ST to distinguish the FFKC group from the normal group using the RF (version 4.6-14) and NN (version 1.44.2) packages. Parameters exported from the Pentacam, PRFI, TBI, and BAD-D were not included in the training. In brief, for the RF model, 500 decision trees were grown and combined to converge the out-of-bag error and improve the prediction performance.[14] For the NN model, an artificial neural network was built on multiple layers of interconnected nodes, including two hidden layers and four hidden neurons, using a supervised learning algorithm.[15]

Validation

The total dataset was randomly divided into a training and a validation set to determine the clinical validity of the two models and their ability to correctly analyze new data: the training data constituted 70% of the total data set and was used to train the models, whereas the remaining data formed the validation set used to evaluate the models' accuracy. The average value of the classification accuracy obtained after executing a 10-fold cross-validation was recorded.

Statistical Analysis

The statistical analysis was performed in SPSS software (version 24; IBM Corporation) and R software (version 4.0.4). The normality of the data was verified using the Shapiro-Wilk test. Descriptive statistics were presented as mean ± standard deviation. For continuous variables, analysis of variance and Kruskal-Wallis H test were conducted to analyze the differences be-

TABLE 1
Basic Demographic Information

Unit	Normal	FFKC	EKC	AKC
Number (OD/OS)	110/113	36/33	35/37	128/130
Age (Years)	22.07 ± 6.32	24.55 ± 5.34	23.25 ± 6.86	22.19 ± 6.18
Gender (M/F)	111/112	34/35	38/34	132/126
Kmax (D)	44.38 ± 1.48	44.35 ± 1.47	46.08 ± 2.41	54.14 ± 7.07
Thinnest pachymetry (μm)	538.64 ± 31.20	528.64 ± 27.62	506.68 ± 29.13	473.71 ± 38.38
IS-Value (D)	0.21 ± 0.66	0.42 ± 0.46	1.52 ± 1.36	4.45 ± 3.25
bIOP (mm Hg)	15.95 ± 2.11	14.76 ± 2.20	14.93 ± 2.10	14.07 ± 2.34
SPA1 (mm Hg/mm)	112.29 ± 17.10	98.64 ± 17.66	93.94 ± 19.26	73.31 ± 21.63
PRFI	0.13 ± 0.11	0.21 ± 0.16	0.66 ± 0.24	0.91 ± 0.19
BADD	1.18 ± 0.65	1.42 ± 0.63	2.94 ± 1.60	7.14 ± 3.79
CBI	0.23 ± 0.27	0.36 ± 0.32	0.56 ± 0.37	0.87 ± 0.27
SSI	0.87 ± 0.12	0.85 ± 0.15	0.85 ± 0.14	0.78 ± 0.16
TBI	0.26 ± 0.19	0.38 ± 0.26	0.87 ± 0.27	0.97 ± 0.13

AKC = advanced keratoconus; BADD = Belin-Ambrósio Enhanced Ectasia Total Deviation Index; bIOP = biomechanically corrected intraocular pressure; CBI = Corvis Biomechanical Index; D = diopters; EKC = early keratoconus; FFKC = forme fruste keratoconus; IS-Value = the difference between average inferior and superior corneal powers 3 mm from the center of the cornea; Kmax = maximum keratometry; OD = right eye; OS = left eye; PRFI = Pentacam Random Forest Index; SPA1 = stiffness parameter at first applanation; SSI = stress-strain index; TBI = Tomographic and Biomechanical Index

tween the four groups, and post-hoc tests were performed with a Bonferroni correction. The 95% CIs were calculated by the Binomial exact test. A *P* value of less than .05 was considered statistically significant for all tests. To determine the optimal cut-off values, sensitivity, and specificity, we used receiver operating characteristic (ROC) curves and area under the curve (AUC) as accuracy measures. Whereas an AUC value of 1.0 indicates perfect discrimination, values of 0.5 or less show that the assessed parameter has no diagnostic ability. Values between 0.5 and 1.0 refer to a significant difference between the distributions of the considered variables in the compared groups. The top 10 AUC values and existing AI parameters' ROC result of all compared groups were taken and sorted from high to low.

RESULTS

Demographics

This retrospective study included 622 eyes of 481 patients for whom the demographic information is shown in **Table 1**, and the parameter distributions and comparisons between groups are shown in **Figure A** (available in the online version of this article). There were significant age differences between groups (*P* = .017), especially between the FFKC and normal groups, and the FFKC and AKC groups (*P* = .004 and .005, respectively). There were no significant differences in sex distribution and right eye/left eye ratios (both *P* > .05) between the groups.

ROC Analysis

The 10 highest-ranked parameters according to their AUC among those obtained from the SD-OCT (marked with ※), Corvis ST, and Pentacam (only PRFI, BAD-D, and TBI related with Pentacam) are shown in **Table 2** for the comparisons of the normal group with the three keratoconus groups. In comparing the normal group with the AKC and EKC groups, the top three AUC ROC ranked parameters were the same in sequence: PRFI (AUC = 0.985 and 0.958, respectively), TBI (AUC = 0.983 and 0.925), and BAD-D (AUC = 0.981 and 0.922). When comparing the normal group with the FFKC group, the best AUC ROC greatly declined, and the top three AUC ROC ranked parameters switched from PRFI, TBI, and BAD-D to the following independent DCR parameters: A2_Deflection_Amp (AUC = 0.761), A2_Deflection_Area (AUC = 0.755), and A2_Deflection_Length (AUC = 0.701).

Feature Selection and Artificial Intelligence Models

The results of feature selection from SD-OCT and Corvis ST are provided in **Table A** (available in the online version of this article). For AI models performance, based on the selected features from SD-OCT and Corvis ST, the RF and NN performed well, which far outperformed the existing clinical parameters PRFI, BAD-D, CBI, and TBI (**Figure 1**).

In detail, the best trained model was based on the RF by combining features from SD-OCT and Corvis ST

TABLE 2
Top 10 AUC Parameters in All Compared Groups

Variable	AUC	95% CI	Cut-off Value	Sensitivity	Specificity
Normal group vs AKC group					
PRFI	0.985	0.974-0.997	0.410	0.982	0.957
TBI	0.983	0.971-0.995	0.595	0.969	0.965
BAD-D	0.981	0.970-0.993	2.350	0.978	0.946
CBI	0.934	0.911-0.957	0.835	0.969	0.821
EPIOverallStdDev ※	0.906	0.879-0.932	2.665	0.834	0.833
Integrated_Radius_mm^{-1}	0.880	0.850-0.911	10.480	0.883	0.755
Max_InverseRadius_mm^{-1}	0.878	0.847-0.909	0.202	0.897	0.743
DA_Ratio_Max_2mm	0.877	0.846-0.908	5.170	0.951	0.685
PachyemtryOverallStdDev ※	0.863	0.831-0.896	22.075	0.848	0.774
EPI5mmSNIT ※	0.851	0.815-0.887	1.625	0.924	0.735
Normal group vs EKC group					
PRFI	0.958	0.927-0.990	0.365	0.960	0.861
TBI	0.925	0.884-0.965	0.750	0.978	0.819
BAD-D	0.922	0.881-0.963	1.875	0.897	0.847
EPI5mmSNIT ※	0.761	0.690-0.832	1.325	0.901	0.597
CBI	0.758	0.687-0.829	0.465	0.798	0.639
EPI5mmSuperiorInferior ※	0.721	0.646-0.797	-0.130	0.767	0.583
Def_Amp_Max_mm	0.719	0.649-0.789	1.066	0.619	0.736
HC_Deformation_Amp_mm	0.719	0.649-0.789	1.066	0.619	0.736
EPI5mmSI ※	0.711	0.632-0.789	0.625	0.870	0.500
DA_Ratio_Max_2mm	0.697	0.627-0.767	4.491	0.596	0.750
Normal group vs FFKC group					
A2_Deflection_Amp_mm	0.761	0.685-0.836	0.134	0.830	0.667
A2_Deflection_Area_mm^{-1}	0.755	0.680-0.831	0.332	0.812	0.667
A2_Deflection_Length_mm	0.701	0.630-0.772	3.006	0.525	0.812
Def_Amp_Max_mm	0.676	0.600-0.752	1.062	0.587	0.710
HC_Deformation_Amp_mm	0.676	0.600-0.752	1.062	0.587	0.710
A2_Deformation_Amp_mm	0.674	0.604-0.744	0.373	0.475	0.783
BAD-D	0.654	0.579-0.729	1.465	0.740	0.580
HC_Deflection_Amp_mm	0.652	0.573-0.732	0.931	0.735	0.522
PRFI	0.652	0.575-0.729	0.165	0.717	0.536
DA_Ratio_Max_2mm	0.648	0.573-0.723	4.400	0.489	0.797
TBI	0.647	0.571-0.723	0.305	0.623	0.609
CBI	0.632	0.556-0.707	0.465	0.798	0.435

AKC = advanced keratoconus; AUC = area under the receiver operating characteristic curve; BAD-D = Belin-Ambrósio Enhanced Ectasia Total Deviation Index; CBI = Corvis Biomechanical Index; EKC = early leratoconus; FFKC = forme fruste keratoconus; PRFI = Pentacam Random Forest Index; TBI = Tomographic and Biomechanical Index

(AUC = 0.99; 88.89% accuracy; 75.00% sensitivity; 94.74% specificity), following the RF-based model by only using Corvis ST features (AUC = 0.92; 90.00% accuracy; 72.22% sensitivity; 94.44% specificity). The performance of NN-based AI models was worse than that of RF-based AI models. However, the NN-based model that combined features from SD-OCT and Corvis ST (AUC = 0.88; 90.12% accuracy; 73.68% sensitivity; 87.10% specificity) was still better than the NN model that used the features from Corvis ST alone (AUC = 0.89; 90.12% accuracy; 63.16% sensitivity; 96.77% specificity).

DISCUSSION

This diagnostic study represents the first attempt to innovative combine diagnostic information from SD-OCT and air-puff devices by AI to enhance clinicians' ability to detect keratoconus and in particular, FFKC, the results were superior to using either of the devices individually.

As expected, the parameters already provided by the Pentacam, the Corvis ST, and the SD-OCT had an excellent ability to distinguish between normal and AKC corneas, reflecting normal clinical practice where they are already used to diagnose AKC using topography or tomography maps. The four most successful parameters in identifying AKC corneas were all AI parameters based on either RF (PRFI and TBI) or logistic regression (CBI and BAD-D), emphasizing the importance of AI-assisted keratoconus diagnosis. The rest of the 10 best performing parameters included the overall standard deviation in epithelial thickness and overall corneal pachymetry thickness (obtained with SD-OCT imaging), which is compatible with the expectation that AKC corneas had undergone a major change in corneal thickness. It is worth noting that SSI, associated with the corneal material's biomechanical properties, did not appear in the list of best classifiers, which may be a confirmation of the idea that keratoconus originates from a localized (rather than a global) decrease in corneal biomechanics.[16]

In comparing the normal and the EKC groups, the three best performing parameters based on AUC were also AI parameters (PRFI, TBI, and BAD-D). Although the AUC values were lower than for the comparison with the AKC group, they still showed an outstanding diagnostic ability (**Table 2**). For the other parameters, the AUC was considerably lower compared to PRFI, TBI, and BAD-D, again confirming the importance of AI parameters. Further, the SD-OCT data showed a regional epithelial remodeling (**Table 2**), reported earlier by Silverman et al,[17] confirming epithelial thickness (ET) redistribution as one of the most critical AS-OCT parameters for detecting early-stage keratoconus.

Finally, for discriminating between the normal and FFKC groups, a dramatic decrease in AUC was observed in all parameters measured, compared with the AKC and EKC analyses. Among the 10 top-scoring AUC parameters, independent DCR parameters replaced AI parameters and occupied dominant positions. These parameters, and particularly the first three parameters (A2_Deflection_Amp, A2_Deflection_Area, and A2_Deflection_Length), were more focused on the second applanation event, and were not independent variables that formed parts of the CBI and TBI. When compared with the DCR parameters, the tomographic parameters derived from OCT were less

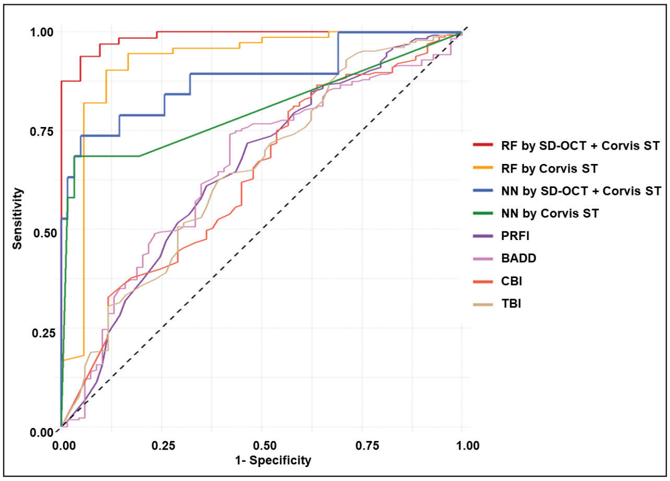

Figure 1. Comparison of four artificial intelligence models and existing comprehensive parameters with receiver operating characteristic (ROC) curve analysis. The Corvis ST is manufactured by Oculus Optikgeräte GmbH. BADD = Belin Ambrósio Enhanced Ectasia Total Deviation Index; CBI = Corvis Biomechanical Index; NN = neural networks; PRFI = Pentacam Random Forest Index; RF = random forest; SD-OCT = spectral-domain optical coherence tomography; TBI = Tomographic and Biomechanical Index

important, suggesting that the biomechanical change that takes place in keratoconus occurs earlier than its morphological change in line with earlier literature.[5] The BAD-D and PRFI, based on corneal tomography, are also relevant morphological parameters with some degree of ability to diagnose FFKC.

A literature review conducted by the authors of the current study identified six recent studies that focused on the detection of EKC (including subclinical keratoconus, EKC, and FFKC) by various AI models (**Table B**, available in the online version of this article). Analysis of these studies identified several limitations. First, the number of EKC eyes included was limited, especially for FFKC eyes; indeed, in three studies[18-20] no FFKC eyes were included. In another study by Xie et al,[21] although a large number of EKC eyes were included, the EKC inclusion criteria were ambiguous. In addition, the inclusion criteria across studies were inconsistent, making it difficult to compare the performance of the AI models used. Finally, no biomechanical information was included, which may have impacted the models' performance. To address these points, the current study included both FFKC and EKC eyes, and had the same inclusion criteria as the existing comprehensive parameters (PRFI, TBI, and CBI). In addition, both corneal structural and biomechanical information obtained from SD-OCT and Corvis ST were considered in the analyses.

Multinomial logistics regression was not applied in the current study because it would delete features to get the minimal-optimal features. However, because multinomial logistics regression cannot optimally handle the relationship between the deleted and the reserved features, this process can result in a decrease in the model's performance. One limitation of this study was the relatively small size of the FFKC group due to the strict inclusion criteria. Because a larger population of patients with FFKC may improve the performance of the AI models, we continue to search for more patients with FFKC for future analyses.

This study confirms that earlier AI diagnostic tools that relied on PRFI, TBI, and BAD-D could accurately diagnose patients as having AKC and EKC, but struggle to diagnose FFKC. In contrast, the current RF implementation combined corneal shape features obtained through SD-OCT with the biomechanical parameters of the air-puff device, which greatly improves health care professionals' ability to diagnose FFKC, leading to an overall improvement in the treatment of patients with keratoconus.

AUTHOR CONTRIBUTIONS

Study concept and design (N-JL); data collection (N-JL, L-LC); analysis and interpretation of data (N-JL, AE, JJR, NH, IMA, MH, DE, CF, CK, FH); writing the manuscript (N-JL); critical revision of the manuscript (N-JL, AE, JJR, NH, IMA, MH, DE, CF, CK, L-LC, FH); statistical expertise (N-JL, AE, JJR, DE, CF); administrative, technical, or material support (AE, JJR, FH); supervision (AE, JJR, IMA, CK, L-LC, FH)

REFERENCES

1. Torres-Netto E, al-otaibi W, Hafezi N, et al. Prevalence of keratoconus in paediatric patients in Riyadh, Saudi Arabia. *Br J Ophthalmol*. 2018;102:bjophthalmol-2017.
2. Li Y, Tan O, Brass R, Weiss JL, Huang D. Corneal epithelial thickness mapping by Fourier-domain optical coherence tomography in normal and keratoconic eyes. *Ophthalmology*. 2012;119(12):2425-2433. https://doi.org/10.1016/j.ophtha.2012.06.023 PMID:22917888
3. Hwang ES, Perez-Straziota CE, Kim SW, Santhiago MR, Randleman JB. Distinguishing highly asymmetric keratoconus eyes using combined Scheimpflug and spectral-domain OCT analysis. *Ophthalmology*. 2018;125(12):1862-1871. https://doi.org/10.1016/j.ophtha.2018.06.020 PMID:30055838
4. Vinciguerra R, Ambrósio R Jr, Elsheikh A, et al. Detection of keratoconus with a new biomechanical index. *J Refract Surg*. 2016;32(12):803-810. https://doi.org/10.3928/1081597X-20160629-01 PMID:27930790
5. Vinciguerra R, Ambrósio R Jr, Roberts CJ, Azzolini C, Vinciguerra P. Biomechanical characterization of subclinical keratoconus without topographic or tomographic abnormalities. *J Refract Surg*. 2017;33(6):399-407. https://doi.org/10.3928/1081597X-20170213-01 PMID:28586501
6. Ambrósio R Jr, Lopes BT, Faria-Correia F, et al. Integration of Scheimpflug-based corneal tomography and biomechanical assessments for enhancing ectasia detection. *J Refract Surg*. 2017;33(7):434-443. https://doi.org/10.3928/1081597X-20170426-02 PMID:28681902
7. El Wardani M, Hashemi K, Aliferis K, Kymionis G. Topographic changes simulating keratoconus in patients with irregular inferior epithelial thickening documented by anterior segment optical coherence tomography. *Clin Ophthalmol*. 2019;13:2103-2110. https://doi.org/10.2147/OPTH.S208101 PMID:31802839
8. Reinstein DZ, Archer TJ, Gobbe M. Corneal epithelial thickness profile in the diagnosis of keratoconus. *J Refract Surg*. 2009;25(7):604-610.
9. Vinciguerra R, Ambrósio RJ, Elsheikh A, et al. Detection of post-laser vision correction ectasia with a new combined biomechanical index. *J Cataract Refract Surg*. 2021;47(10):1314-1318. doi:10.1097/j.jcrs.0000000000000629
10. Lu NJ, Chen D, Cui LL, Wang L, Chen SH, Wang QM. Repeatability of cornea and sublayer thickness measurements using optical coherence tomography in corneas of anomalous refractive status. *J Refract Surg*. 2019;35(9):600-605. https://doi.org/10.3928/1081597X-20190806-03
11. Roberts CJ, Mahmoud AM, Bons JP, et al. Introduction of two novel stiffness parameters and interpretation of air puff-induced biomechanical deformation parameters with a dynamic Scheimpflug analyzer. *J Refract Surg*. 2017;33(4):266-273. https://doi.org/10.3928/1081597X-20161221-03
12. Kursa MB, Rudnicki WR. Feature selection with the Boruta package. *J Stat Softw*. 2010;36(11):1-13. https://doi.org/10.18637/jss.v036.i11
13. Kohavi R, John GH. Wrappers for feature subset selection. *Artif Intell*. 1997;97(1-2):273-324. https://doi.org/10.1016/S0004-3702(97)00043-X
14. Liaw A, Wiener M. *Classification and Regression by RandomForest*. Forest; 2001:23.
15. Günther F, Fritsch S. neuralnet: Training of Neural Networks. *The R Journal*. 2010;2/1:30-38.
16. Roberts CJ, Dupps WJ Jr. Biomechanics of corneal ectasia and biomechanical treatments. *J Cataract Refract Surg*. 2014;40(6):991-998. https://doi.org/10.1016/j.jcrs.2014.04.013 PMID:24774009
17. Silverman RH, Urs R, Roychoudhury A, Archer TJ, Gobbe M, Reinstein DZ. Epithelial remodeling as basis for machine-based identification of keratoconus. *Invest Ophthalmol Vis Sci*. 2014;55(3):1580-1587. https://doi.org/10.1167/iovs.13-12578 PMID:24557351
18. Cao K, Verspoor K, Sahebjada S, Baird PN. Evaluating the performance of various machine learning algorithms to detect subclinical keratoconus. *Transl Vis Sci Technol*. 2020;9(2):24. https://doi.org/10.1167/tvst.9.2.24 PMID:32818085
19. Shi C, Wang M, Zhu T, et al. Machine learning helps improve diagnostic ability of subclinical keratoconus using Scheimpflug and OCT imaging modalities. *Eye Vis (Lond)*. 2020;7(1):48. https://doi.org/10.1186/s40662-020-00213-3 PMID:32974414
20. Kuo B-I, Chang W-Y, Liao T-S, et al. Keratoconus screening based on deep learning approach of corneal topography. *Transl Vis Sci Technol*. 2020;9(2):53-53. https://doi.org/10.1167/tvst.9.2.53 PMID:33062398
21. Xie Y, Zhao L, Yang X, et al. Screening candidates for refractive surgery with corneal tomographic-based deep learning. *JAMA Ophthalmol*. 2020;138(5):519-526. https://doi.org/10.1001/jamaophthalmol.2020.0507 PMID:32215587
22. Pavlatos E, Chen S, Yang Y, Wang Q, Huang D, Li Y. A coincident thinning index for keratoconus identification using OCT pachymetry and epithelial thickness maps. *J Refract Surg*. 2020;36(11):757-765. https://doi.org/10.3928/1081597X-20200925-01
23. Toprak I, Vega A, Alió Del Barrio JL, Espla E, Cavas F, Alió JL. Diagnostic value of corneal epithelial and stromal thickness distribution profiles in forme fruste keratoconus and subclinical keratoconus. *Cornea*. 2021;40(1):61-72. https://doi.org/10.1097/ICO.0000000000002435 PMID:32769675

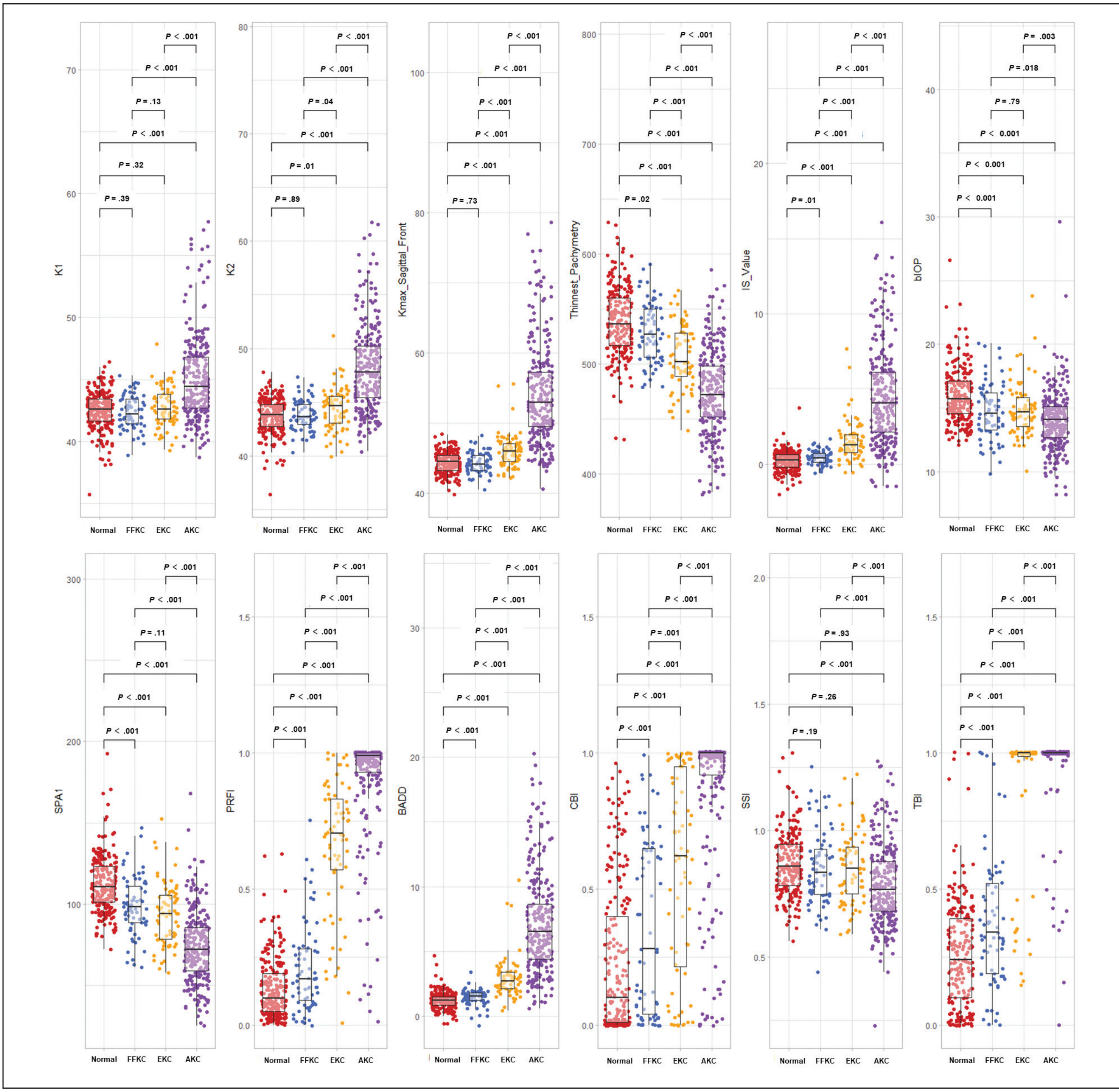

Figure A. The distributions and comparisons of patients' demographic characteristics among all groups. BADD = Belin Ambrósio Enhanced Ectasia Total Deviation Index; bIOP = biomechanically corrected intraocular pressure; CBI = Corvis Biomechanical Index; IS-Value = the difference between average inferior and superior corneal powers 3 mm from the center of the cornea; PRFI = Pentacam Random Forest Index; SPA1 = stiffness parameter at first applanation; SSI = stress-strain index; TBI = Tomographic and Biomechanical Index

TABLE A

Selected Features From OCT and Corvis ST

Parameter	Mean Imp	Minlmp-Maxlmp	Norm Hits	Decision
Selected features from OCT				
EPI2mm	7.56	2.33-9.76	0.97	Confirmed
EPI5mmIT	6.56	1.35-8.54	0.95	Confirmed
EPI7mmIT	6.67	1.55-9.34	0.94	Confirmed
EPI9mmIT	6.18	0.84-8.73	0.94	Confirmed
EPI5mmT	5.31	2.03-7.45	0.91	Confirmed
EPI5mmIN	4.92	1.36-7.18	0.88	Confirmed
Pachymetry9mmS	5.00	0.63-7.62	0.88	Confirmed
EPI7mmI	5.20	-0.09-7.63	0.87	Confirmed
EPI9mmT	4.43	0.57-6.89	0.83	Confirmed
EPIOverallI.2.7mm	4.50	0.55-7.67	0.82	Confirmed
EPI5mmN	4.48	0.69-6.73	0.81	Confirmed
EPI7mmT	4.42	0.46-7	0.81	Confirmed
EPI5mmSN	4.37	0.19-7.2	0.80	Confirmed
EPI5mmI	4.48	1.2-6.78	0.80	Confirmed
Stroma9mmS	4.26	1.32-6.91	0.77	Confirmed
EPI9mmN	4.09	-0.67-6.46	0.72	Confirmed
Stroma5mmST.IN	3.59	0.3-6.14	0.65	Confirmed
Pachymetry9mmSN	3.46	0.09-5.79	0.60	Tentative
EPI9mmST.IN	3.39	0.38-6.02	0.59	Tentative
EPI5mmST.IN	3.25	0.17-6.3	0.58	Tentative
Stroma9mmSN	3.22	0.14-6.06	0.55	Tentative
EPI9mmST	3.01	-0.23-5.59	0.49	Tentative
Pachymetry5mmST.IN	2.81	-0.25-6.27	0.45	Tentative
Selected features from Corvis ST				
A2_Velocity_m_s	24.59	18.97-27.53	1	Confirmed
A2_Time_ms	18.29	15.1-20.63	1	Confirmed
A1_Time_ms	10.85	8.75-12.61	1	Confirmed
A2_Deflection_Amp_mm	10.86	8.77-12.63	1	Confirmed
IOP_mmHg	8.52	6.28-10.93	1	Confirmed
SPA1	8.45	5.42-10.38	1	Confirmed
Whole_Eye_Movement_Max_ms	8.06	5.11-9.85	1	Confirmed
A2_Deflection_Area_mm1	10.11	7.38-12.16	1	Confirmed
bIOP	7.65	5.14-9.61	1	Confirmed
HC_Deformation_Amp_mm	5.36	3.34-7.4	0.97	Confirmed
Def_Amp_Max_mm	5.35	3.01-7.55	0.97	Confirmed
HC_Deflection_Length_mm	4.67	1.67-6.99	0.91	Confirmed
A2_dArc_Length_mm	4.67	1.77-7.31	0.91	Confirmed
Peak_Dist_mm	4.39	1.95-6.82	0.89	Confirmed
Max_InverseRadius_mm1	4.50	1.29-6.78	0.86	Confirmed
Radius_mm	3.93	0.9-6.23	0.79	Confirmed
A2_Deflection_Length_mm	3.87	0.71-6.21	0.79	Confirmed
HC_Deflection_Amp_mm	3.52	1.63-5.47	0.77	Confirmed
HC_Deflection_Area_mm1	3.46	1.77-6.8	0.74	Confirmed
DA_Ratio_Max_2mm	3.30	0.92-5.8	0.66	Confirmed
Deflection_Amp_Max_mm	3.07	0.77-5.19	0.63	Confirmed
Integrated_Radius_mm1	3.10	0.87-5.18	0.62	Confirmed
A2_Deformation_Amp_mm	3.06	0.52-5.69	0.61	Confirmed
A1_Velocity_m_s	2.95	-0.11-5.26	0.58	Tentative
HC_dArc_Length_mm	2.58	-0.13-4.76	0.45	Tentative

bIOP = biomechanically corrected intraocular pressire; Mean Imp = mean importance; Minlmp-Maxlmp = arrangement between minimum importance and maximum importance; Norm Hits = fraction of random forest runs; OCT = optical coherence tomography; SPA1 = stiffness parameter at first applanation
The Corvis ST is manufactured by Oculus Optikgeräte GmbH.

TABLE B

Recent Studies Using AI to Differentiate Subclinical KC and FFKC From Normal Eyes

Study	Devices (Analyzed Form)	Eyes Included	Algorithm Used	Performance
Cao et al, 2020[18]	Pentacam (parameters)	49 subclinical KC eyes, 39 control eyes	Eight different algorithms	Highest AUC = 0.97 with 89% precision by RF
Shi et al, 2020[19]	Combine Pentacam and laboratory UHR-OCT (parameters)	38 KC eyes, 33 subclinical KC eyes, and 50 normal eyes	NN	Highest AUC = 0.93, 93% precision for subclinical KC eyes, and 99% precision for KC eyes by NN
Kuo et al, 2020[20]	Pentacam (images)	170 KC eyes, 28 subclinical KC eyes, and 156 normal eyes	Three different convolutional NN models	Highest AUC = 0.995 by ResNet152 model
Xie et al, 2020[21]	Pentacam (images)	389 KC eyes, 202 EKC eyes, and 1,368 normal eyes	Convolutional NN	AUC = 0.996 with 92% sensitivity, and 99.1% specificity for EKC
Pavlatos et al, 2020[22]	Optovue SD-OCT (parameters)	91 KC eyes, 16 subclinical KC eyes, 26 FFKC eyes, and 82 normal eyes	Repeated five-fold cross-validation	56% sensitivity for FFKC and 100% sensitivity for subclinical KC
Toprak et al, 2021[23]	MS-39 SD-OCT (parameters)	50 subclinical KC eyes, 27 FFKC eyes, and 66 normal eyes	Binary logistic regression	75% sensitivity and 94.3% specificity for FFKC

AI = artificial intelligence; AUC = the area under the curve; EKC = early keratoconus; FFKC = forme fruste keratoconus; KC = keratoconus; NN = neutral network; OCT = optical coherence tomography; RF = random forest; SD-OCT = spectral-domain optical coherence tomography; UHR-OCT = ultra-high-resolution OCT
The Pentacam is manufactured by Oculus Optikgeräte GmbH.